

ActInf GuestStream 054.1 ~ A Gómez-Emilsson & C Percy ~ Electromagnetic Field Topology Consciousness

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foreign this is active inference live stream guest stream number 54.1 it's August 25th 2023. today we're going to be hearing about electromagnetic field topology as a solution to the boundary problem of Consciousness this is going to be a co-presentation followed by a discussion with our guests here so I will leave it to everyone to introduce themselves and present and discuss however they prefer and thank you again for joining please thank you so much uh for having us Daniel uh yeah I mean there is a co-founder of quality of research institute and uh yeah uh I'm here with a Chris and elieber Chris Cloud release paper and and the Lieber volunteer and uh Wiki extraordinaire for qri but yeah go ahead uh Chris and Hillary yeah yeah no no no no more to say that that's plenty let's let's get into the detail yeah I'm good working with qri and a lot into active influence so I'm and I'm trying to unify with active influence a lot and yeah beautiful uh yeah I mean essentially yeah next next slide essentially our um outline of this presentation is gonna essentially walk you through some of the key ideas of the paper um and kind of like trying to illustrate them or making them a little bit more intuitive um fairly high level although we will actually present kind of the the core the core argument um the main thing to to to point out here uh next slide is that the genetic theories of Consciousness you know I guess like 20 years ago they sounded like a total total crackpot type yeah I've still got a good connection so we can give him a moment or two yeah and if not I'll I'll pick up from what was the crackpot vehicle next um sorry unders please repeat from crackpot oh yeah oh my God yeah uh over the last 20 years however there's been an explosion of interest in electromagnetic theories of Consciousness and serious peer-reviewed Publications about them especially the last 10 and then five years kind of like a variable exponential explosion in terms of a number of like serious articles published and one of the very very kind of like key amazing kind of like contributions that something like genetic magnetic Theory Of Consciousness can offer to the overall ecosystem of philosophy of mind

is that it provides a way to solve question for how information can be combined I mean essentially when you have information especially distributed how is it possible that they can be simultaneously part of some kind of like unified whole and electromagnetic Theory Of Consciousness are just very proud to say well the electromagnetic field is ontologically unitary it's already unified so there's not much of a problem here um however you know once you kind of like have this this uh Unity uh then the question becomes well but how do you actually draw a boundary around it so actually you just traded one problem for another so that's the guy yeah dig into the details there cool should I take us through the literature reviewers context for this and then Andrews will come back to kind of give you the the solution that we think can fit with electromagnetic fields so yeah the boundary problem let's so binding problem you heard just now into a first approximation that is whatever you think might make up Consciousness maybe it's neuronspiring maybe it's some sort of magical qualier at the bottom of the universe whatever you think it is these small things need to be bound together to ex to reflect the complex phenomenology we have we see lots of things all at once right now I'm I'm smelling the flowers outside I've got the laptop screen in front of me there's lots of stuff happening all at once in a unified way so that's the binding problem we won't do more on that here we'll sort of assume some familiarity with it but then the boundary problem is well once you've got a mechanism that creates The Binding between these lower level entities why would that mechanism ever stop because clearly our awareness our phenomenology doesn't go on forever we don't know what's happening in Alpha Century our awareness is not unbounded and as soon as you realize that you need boundaries around something a number of things suddenly don't quite make sense so Owen Schrodinger or physics Fame flagged this up as one of the things that he felt was completely unresolved so despite his his kind of vast intellect and his contributions elsewhere he felt that this puzzle was one of the things that he described as when he turns his mind to it everything just evaporates none of it none of it makes sense anymore he can't get a grip on the problem and he described the problem as you can see on the screen he was saying why is it that my identity is at the intermediate level this kind of roughly human-shaped body and you can say okay well mostly mean the central nervous system but either way this this broad entity has lots of smaller things inside it cells organs human bodies and lots of things above it cities states solar systems universes and so on why is our self-consciousness always at this always at this one level and if you describe that as a problem to somebody who's never heard of it before or never thought of it before they may this happened to me just just last Friday they might say well it's obviously just like the body stops like the

skull stops here or like my arm stops here like of of course it's not going out for a century why would it and that's it's a very intuitive immediate human reaction but you probe it a little bit as Schrodinger did and it kind of falls apart because if you're a you're on firing somewhere in the central nervous system and you're being initiated by a neuron in the peripheral nervous system well how do you know to stop binding how do you know that there's a boundary that stops at the edge of the central nervous system or if you're if your fingers are holding onto a tool and you're getting feedback from that tool you're driving a car it can feel like you're part of that car how does where does the system know where the boundary is because most of these things that we observe have only Observer dependent boundaries if you've got a wheel in a car rolling along the road from one perspective it looks like the wheel is separate from the road but if you're a tiny ant or something and the car is parked these things all may feel like they're one object and this was when when Schrodinger turned his mind to this it puzzled him so this let's give him the benefit of the doubt that's let's say that there's something real to be puzzled through this issue applies both to the example I started with around building up in small units but also if you're a cosmo pan psychist you believe that there is one universal Consciousness and we are like a god playing with its puppets in that case you still have the same question because you're still demarcating boundaries you just get the mirror image we'll mostly focus on the idea of expanding up from small things because that's closer to the Orthodoxy but there's no you can't solve this problem magically by by declaring a single consciousness okay so that's that's what that say much more about this he just said that this was a this was a puzzle the first person we found that really formalized this and gave it the name boundary problem was Rosenberg and it was a number of pieces late 90s early 2000s and he uses the phrase sitter and cryptus but it's exactly the same idea as Schrodinger had one of them is well why aren't there lots of small things I think that's I think that might have been the Silla that might have been the the the nine-headed monster and then on the other hand what about the things that we are nested within and he called that the cryptos and he said it's very hard to pause it a genuinely robust mechanism that would stop it in both directions you can imagine a reason why lots of small things get bound together but why would the binding stop and yeah his view in framing this problem is that no one had solved it winding on to Anders and I chatting about this last year and building on some of qri's foundational work in this area we wanted to know well what's happened since Rosenberg framed this and schroding was so puzzled by it like surely someone's come up with a few solutions to it so Q standards database driven literature review and you've got the details there that fairly standard stuff what we what we looked

looking across the scopus database we only found five papers that seem to be relevant about the boundary problem compared to 92 on The Binding and 47 on the combination problem so some of these related problems and in our intro we kind of went straight from binding to boundary they are so you know if you're thinking about one you really should be thinking about the other well that is not reflected in the literature we did a bit more puzzling around people don't seem to be referring to this by other names or when they are they haven't solved it either um so it felt that there was something worth going out here and this is what what initiated the paper what we did find though in partly in those five papers and partly in looking around we do find families of possible solutions and there's no clear winner but maybe philosophy rarely has a clear winner for anything so maybe that's an unreasonable expectation but there's definitely open questions for each class of theories that have been put out there to solve this problem so the first of these Loosely termed algorithmic so computational causal structure theories I'm just trying to summarize them in into one term some of the examples there IIT of course very well known uh Global workspace Theory as well they will tend to say that things bind together when they're in a particular pattern so you might imagine a database that references itself or the working memory of a computer or as soon as things are somehow linked together in a causal sense in IIT there's a particular type of pattern but nonetheless it's an abstract informational relationship those papers don't tend to be super explicit about defining boundaries normally they let's say IIT will normally say here's our system and now let's ask within that system where is the highest Bay but they sort of just say take the boundaries on that says no one has said this take the whole universe as a system that the math is already virtually impossible at even even toy systems but reading between the lines or reading some of the other papers around it there's something about if you had a high fly area surrounded by lots of Lo-Fi areas well it's natural that it's the it's the bit that's high and there's this principle of exclusion that says you take the highest one in a system maybe you have to specify a system before it makes sense to ask that and if you took the highest bit out somehow you just remove some of those causal interconnections that create that that highness whatever's left will have a slightly lower bit of Phi and that is now where the the the sort of lockers of Consciousness resides this may well work kind of the maths would need working out I think Andrews and I are initially skeptical um but because as best as we can see no one's really created complicated maps and discussed what would happen when you've got different mixtures of densities that are shifting over time what stops it from being arbitrary certainly there are there are questions to be had and we can come back to this a bit if we want to talk

about it further but surprise it to say it's not a guaranteed solution and there's one critique that we cite resonance theories a link between those and electromagnetic fields now as well in the the main progenitor of these theories Tom Hunt would say that the the main resonance which he sees binding happening together in human consciousness is enfields so there are there are links between these as well and maybe the informational patterns are being encoded in resonance or quantum entanglement or even feel so there can be links between these as well the resonance theory that says The Binding happens when things are in the same frequency simple enough or when they are close enough and that's becomes a little bit more debatable but there is you know what does that really mean the boundaries there is a conceptual solution for this which is the slowest shared resonance in a system but many of the same issues applies with as with the algorithmic um but again if experimented it it may be possible to create create one that works there uh quantum theories entanglement is what binds it together at least in some of them so what you know what causes two experiences a blue cut and the shape of the cup to be bound together into one well it's that the blueness and the cutness I'm obviously talking at a very high level of abstraction but those are somehow Quantum entangled with each other that would indeed create to the best of our physics knowledge today a kind of ontologically real binding and that's that's quite an attractive part of the solution well what would stop those boundaries there is the monogamy principle things can only be maximally entangled up to a certain point but there's a very very open question about whether that sort of thing would would enable the kind of macro phenomenology we have but in the paper we say all of these are worth exploring like let's let's keep investigating and then there's the one I I started with an introduction like maybe it's the boundary of the central nervous system but hopefully even a small amount of scrutiny causes that one to uh to collapse and most of these explicitly say that more work is needed more math is needed and certainly that will be true of what we're presenting here as well what we want to do is add a new one to this list and the the inspiration for this is that well if the EM field accounts presents a solution to The Binding problem they certainly believe they do there's lots of literature on it then maybe they can also provide a solution to the boundary problem given how closely related they are when you read those accounts they don't quite have it nailed yet the most common explanation they give is a kind of inverse Square law rule where well the strength of the field just becomes so small so quickly you know by the time you're sent to meter out from the thalamus or by the time you're at the edges of the skull these fields have just evaporated into they decohered they're indistinguishable from the neural noise all around us that might be okay if you're okay with a

fuzzy boundary and we'll come on to that in a sec I always motivate a phase transition a bit like quantum entanglement or a bit like the exact Resonance of resonance theories but those phase transitions are pretty speculative as well so the solution where we will come to an unusual outline is a different aspect of Enfield it's not fuzzy boundaries and it's not a vague phase transition it's an entirely different topological solution which we argue can solve the problems what we think is worth doing though is to make that boundary problem more explicit what you've heard both from Schrodinger and from Rosenberg is this idea of and like why aren't the cells inside you conscious or if they are conscious why don't we ever experience that and what again going up what about cities and families and things all group charts or football stadiums that sort of stuff oh and there is a there is a scholar who argues based on resonance theory that actually when we are in a group chanting or a group dancing situation our Consciousness does in fact move to that level as well and that's it's speculative but you know there are there's much to be thought about so to make this problem more precise we've got five and I'll just give you the the one or two sentences on each the hard boundary problem is this sense that a fuzzy boundary probably isn't good enough and that's it's kind of explain a bit more why if you have something like a wave or the example of the wheel I gave before touching the ground or indeed any connection between any physical atoms the closer you look to it the less obvious it is where it stops the boundaries of my skull may feel very precise to me but if there were an alien the size of a solar system their ability to perceive this would most likely just blurs imperceptibly into its surroundings like these things look feel hard but they're not really hard at the edges of our awareness there's this sense that things I might feel funny but at some point it just sort of stops might feel like it might feel Infamous there are states of mind where we we can feel an infinite awareness but probably bring that further it doesn't truly Encompass everything and go on forever at this Edge our assertion is that sometimes this feels like a hard boundary and if it ever sometimes feels like a hard boundary then you need some mechanism that is capable of a hard Edge in a way that these physical phenomena typically ask so that's the heart boundary definitely happy to like drill into more of any of this or any of the other five if they're not making enough sense in the Q a so lower levels we've covered that that's I mean the the great example of this is um we know that when we go to sleep some part of our Consciousness at least a bit that we remember switches off yet the brain remains active the lower levels boundary problem could be reinterpreted as saying why isn't why doesn't our awareness just shift to another part of the brain at that point it certainly doesn't seem to most of the time higher level boundary problem we've had that from the beginning

that's why doesn't it go up maybe it does sometimes certainly doesn't normally the private boundary problem so this is now rather than going down into ourselves or up into the universe in which we are nested this is more take two of those boundaries me and you me and one other person why can we not Bridge into each other's minds at least not routinely those who claim to do so have struggled to replicate it and get it into the mainstream so this is privacy there appears to be some limitation that stops these mechanisms from connecting with each other and yet if it is slower shared resonance or if it is highest informational Phi in IIT why is it not possible to construct a system in which the highest buyers now interconnected one and our Consciousness is collapsed maybe it is possible and that's just a technical Endeavor we haven't yet achieved by the way there's something that's challenging about it either as a technical barrier or as a philosophical one the last one is the temporal boundary problem because everything we've said so far is in space but the kind of entities and experiences we're talking about are not zero zero time space-based activities they they have they have a temporal persistence so what is it that puts boundaries around a micro moment of experience what is it that enables us to experience the end of a note and still retain that idea of the start of the notes and experience them together and how does this extend through through border experiences of the self so those are the five puzzles hopefully that's enough of a qualitative overview so that you know what we're trying to solve at least and if I hand back over to Andrews now he will give you an entirely new solution to these puzzles yes uh next slide they're going a little bit uh yeah kind of like a little bit broader too in that uh what it creates talked about is kind of the criteria for the boundary problem kind of like breaking down the boundary problem into sub components and then there's like also like more broadly like things that you would want out of a theory of Consciousness and so like the solution to the boundary problem ideally will also in a sense like satisfy these constraints uh and these constraints are framing variants not only phenomenalism and no strong emergence next slide so framing variants essentially can be Recon Lawrence invariance uh the fact that like whatever structure is giving rise to the boundary shouldn't in a way be frame specific shouldn't just depend on like how you're looking at an object it should be kind of like an intrinsic objective property of the object that if you move it around it it should in principle you know be changing in that way next slide for those who are not familiar with the term this is the idea that for a particular conscious physical system that corresponds and experience but that there's no causal influence from that conscious experience back into the and there's like various ways of kind of uh trying to ground the question of hey why why is Zippy phenomenalism not the case um there's kind of again

differences in emphasis by each of these questions but they're roughly pointing in the same direction like one important question is how is it possible that we can talk in detail about our experience but the way we actually framed it in the paper that we find the most compelling is how was Evolution able to recruit you know coherent globally bound States if they're not causally significant right like if they're not causally useful it would just be an extraordinary coincidence that it just happens to happen simultaneously along with something that seems to be computationally significant and you know Encompass a whole lot of information at the same time uh next slide uh and then we also have no strong emergence uh there's these yeah quote from David that says the lagrangian or the standard of the standard model does not break down inside one's own skull meaning we want the laws of physics to apply universally to whatever explanation space would come up with then we don't want to introduce the special cases or say like oh there's like a special different field in the brain or anything of the sort we want to have a homogeneous set of physics that you know as simple as it gets as mainstream As It Gets In terms of uh appealing to the standard physics uh next slide uh so yeah the overarching solution and explanation space that the thing is electromagnetic field topology so first of all I will walk you through in what way these explanation space satisfies the kind of like three constraints that I provided and then increase is going to explain how it addresses each of the sub problems so so yeah uh next one yeah if you actually kind of look at this uh amazing reconstructions of the electromagnetic field of Stellar bodies you would see that nothing trivial about it right because like all of this incredible complexity and richness uh in structure uh a lot of it is very counterintuitive you know not something that you would see with a naked eye you see kind of this you require um superimposing a lot of pictures from the different regions of the spectrum to kind of like get a picture of what's going on in the electromagnetic field and yeah it has kind of this elongated weird plasma tubes and you know circular structures and loops and and so on uh so it's a you know very structurally rich environment uh next slide and you know the way in which you know something like the topology of that field will satisfy the criteria that weight line is as follows so if you look at the electromagnetic field of the sun sometimes this phenomena called magnetic reconnection happens and this is actually responsible for solar flares and kernel Mass ejections it has to do with when plasma tubes that are electromagnetic confinement structures uh touch with each other and so they can you know open up and spray plasma into the the outer space um the key thing to point uh about here is that this is actually a topological change to the field is it's not just any kind of physical change you know when you have a you know solar flare kernel mass

ejection that is the effect of a top macroscopic gigantic topological transformation what this slide is trying to get at is that whether a system experienced magnetic reconnection or not will not depend on your frame of reference and the very simple reason is that lowering transforms simply squeeze and stretch but they never actually you know reconnect or cut or um you know add any kind of like topological changes in other words a Lawrence transform will leave a topological event untouched so in that sense okay like topology as an explanation space for why um why this is framing variant it definitely fits the bill next slide I importantly to you in the brain you know this um we also can tackle the question of like in what way the field is not Epi phenomenal uh there's a lot of research on local field Potentials in the which are yeah microscopic electric oscillations um and they do seem to actually have uh causal effects on the activity of neurons in other words uh people might believe in a sense that hey brain waves are kind of just a reflection of neural activity but the thing that something like transcranial magnetic stimulation makes clear is that on top of that they actually feed back into the probability for neurons to fire especially the probability for nearest to fire in synchrony or in a in a state of coherence in other words the actual causal structure of electromagnetic field and how it connects to neural activity is this uh dual causal Arrow where neural activity influences local field potentials neural activity influences activity via synapse yeah we've been the victim of the connection again let's hope it comes back as quickly as as I did last time hmm yep there's a lot to say but I won't actually you got some some hard questions written down as we've been we've been taking you through it yeah just enough people who are listening in the discussion we'll uh look at some questions so please feel free to write anything personally I'm I'm curious whether the awareness is going to be the the light of the Sun the heat of the Sun the ejection the loop structure the birth and death of the star yeah yeah I mean if that's if the idea that you'll you'll see here stands then you've got the idea that a a closed topological pocket of electromagnetic field that's that's where awareness can take place that's where a first person perspective that kind of a lockers can exist but what it's aware of well that's an entirely different question and if there are Pockets like that I mean we're not asserting in this paper that the song has these it's such a complex entity it's very possible but that's you know that is a different but if to the extent they are there what on Earth would be experienced within it that's a that's a different thing the content it's not just for YouTubers it's also for conscious systems but yeah you've bound it all together you've delimited a boundary topologically and then about what and so what yeah yeah it's me let me pick up a bit while we're while we're waiting for him to comment yeah this is and there's really growing kind of Neuroscience and biological

evidence for downward causation from electromagnetic fields in the brain and all we really do in this paper is acknowledge that research literature exists and say and therefore it's possible for fields to be non-epi phenomenal we don't get into the specifics of it but they've been shown in memory formation they've been shown in the old factory bulb for your smells they've appeared all over the place so there's a ah Andre's your bath yeah it's just making the case for non-epi phenomenal feels fantastic beautiful yeah sorry about that hopefully it doesn't happen again okay next slide yeah and uh well think about like the immense causal power that a topological change in the field has in cosmological terms you know like this billions of tons of plasma being ejected just because of you know otherwise like every phenomenal magnetic reconnection is like no no actually you know topological Transformations are like really really significant at a massive massive scale uh next slide yeah uh next one yeah just sorry kind of how ridiculously powerful this can be yeah and uh importantly too um the actual fact you get an enclosure with a topological transformation in a medium also gives you this capacity to essentially have holistic Behavior Uh this is one of the things that to me was kind of the most exciting about this explanation space and like the moment you can close the field lines or you know do a transformation waves will be confined some kinds of waves can be confined within a topological pocket when you have confinement of waves then you have resonant modes and when you have resonant modes then you have these very fantastic effect where the whole shape of an object essentially is expressed in the way every part of the object is vibrating in in which case yeah essentially you have this reason why Evolution would recruit you know these topological objects is because it actually allows organisms to behave as a unit so I think yeah that might be a very key like piece of the puzzle for for a pre phenomenalism uh next slide uh these are just like yeah kind of like some sketches of some of the kinds of boundaries that in principle you may have in the electromagnetic field and what kind of uh you know binding in Consciousness they might correspond to um but you know you don't have to you know just speculate to the really exciting thing here is that you can actually play with simulations of the electromagnetic field to try to make sense of the topological Transformations that happen in it uh so the next following slides uh we can just go through them fairly quickly are essentially resonant modes of an empty cavity electromagnetic resonance modes of an empty Cube cavity where the left side is the electric field and the right side is the magnetic field there are websites where you can check this is uh called filestat.com embox and the thing that it kind of like really neatly shows is how the magnetic field sometimes get this uh close looping structure uh next slide yeah just go through through them maybe like five seconds for each of them um and

uh it also allows you to essentially look at what happens when you stack some of these resonant modes and some combinations linear combinations of these resonant modes create intricate edit adding uh headings and loops and twists that loop around and create this topological boundary very fleeting it has to do with like the you know the frequency of oscillation of the electric field you know how many times per second you're going to get this looping structure but the fact that yeah you can show there's some topologically closed objects uh that appear there is quite remarkable and overall the idea here would be that well something like this is happening in the brain and the brain is actually recruiting these looping architectures all right I'll hand it over to Chris cool thank you very much yeah so we've seen that electromagnetic fields can form closed loops and people have used that some of the sites in the paper around people trapping knots of light and it just kind of really remarkable things happening at the edge of sort of room temperature State physics really incredible stuff we know that in principle these fields are capable of downward causation um in massive context like the Sun but also in very real human brain settings so conceptually and we know that topologies in general are frame and variants so we've got we've got those things in place we've got boundaries that satisfy those those various features that we wanted we don't know exactly how they occur in the human brain all we're presenting is a conceptual solution that those things are possible but what we do say is well what would what would such a solution look like to those five uh boundary uh problems that we just described so taking them one by one um or rather yeah taking them as a as a group let's first say what's in such a theory the first person perspective that kind of phenomenal awareness the what it's likeness that we experience in a very normal um everyday sense what could that reflect what could that correspond to in such a model and there are others but we'll give one example and use that to walk through the problems so we would say that once you've got a closed topology and it may well require the time dimension for it to be properly ontologically closed and at that point em energy of that specific Spectra the same Spectra that's making up the boundary can't pass through it or not that's that's the whole point about the closed loop other things might pass through it the sound wave might pass through it but nonetheless for the entities that we're interested in it is closed to itself and that's where we get that's where we can get this hard there may be lots of these being created in the brain the brain is absolutely chock-a-block with different bits of em generating underlying phenomena so it's certainly plausible that some of them are closed we don't know for this to be true there must be at least one but it's plausible that if there's one there's probably several so you've got several of these by definition they are

non-overlapping they the boundaries cannot overlap but they could well be multiple at the same time one at the front run at the back could be all over the place it's probable though that only a subset would be continually recreated in a stable manner because the fact that these brain these fields can have downward causation they can do useful things some of the sites in the paper about information processing point to some of the non-linearities you can achieve through wave and field based resonant mode based computations which may well have benefits from from an information process in perspective so in some cases these will be continually recreated only one of them because they cannot overlap only one of them can enclose let's call it an immediate memory model as a conceptual Point not not committing to what exactly that might mean we suspect these wouldn't last very long and they probably don't extend over the whole grain but who knows regardless it would vary from period to period other field theories might have other ways of recruiting this concept for solving the boundary problem but let's walk this one through and see what it looks like as a response to those five things so what we're saying is this closed topological pocket that is kind of axiomatically the same as a first person awareness when you've got that you've got something that is a is capable of both Binding Together those micro bits of experience and presenting them within a boundary so they have some unified ontology one by one and I think this is our last the hard boundary well that comes from what we've been talking about energy of the same Spectra can't pass through that the closed loop so that that happens quite naturally a bit like with the field solution to binding this is something you almost get for free as a result of using that low levels and higher levels well it because only one of them because of the non-overlapping point because only one of them can include the immediate memory model which is crucial for the next bits then you can't have lower levels and higher levels there may well be mined dust at those levels which is the kind of lovely evocative concept maybe there may well be flashes of tiny bits of awareness that's kind of they never repeat the patterns just gone forever there's nothing that links it into future versions of itself um just on the universe may be replete with these things it's certainly got plenty of electrons in it why not plenty of these things who knows but there's only one closed pocket that maybe I'll come back to four that can enclose this immediate memory module and that's crucial because it's what creates a temporal boundary that in any possible sense could reflect what we experienced day to day because these tiny if you think of the these 4D Pockets if if one pocket were to last long enough to Encompass say a whole day's worth of your experience it's almost impossible but it's a very enough without breaking to actually contain all the types of complex phenomena we

experienced during the day which is why we think it's much more likely they persist for a short period of time but nonetheless our experiences or things persisting for longer now the micro level persistence why is it non-zero because if something is infinitesimally small it probably doesn't help us if something is the equivalent of a geometric point no matter how many of them you put together in a physical setting you're not going to create a line we want some non-zero duration well we've got that for free because the closed Pockets has this for the ontological presence that's fine how would you get but it's but it may be really really short far shorter than we experience even this sentence so what ties that together well because it encloses each millisecond to millisecond it recreates each one is a different piece of awareness they are not in any way ontologically linked to each other they are each separate but because each one encloses the memory module which is just an information processing thing you can encode memory and pen and paper there's nothing magical about this by referencing it it knows what was just there before and that can create you can it may we could call it an illusion if you want whether that matters is a much more philosophical point but you create this pseudo-time Arrow you have unified insights because that's what Fields do they unify the insights in all the things that make it up no information is lost it is all bound together so now you've got the recent pasts and then you've got the bit before that from the previous pocket and a bit before that and a bit before that so it's all bound into each each individual presence as short as it may be it has the sense that it's something bigger now in terms of the macro level days years between interruptions this is one where where we're far more philosophically relaxed it it may well be that there is there is no kind of profound sense in which these things are the same entity it's maybe a ship of Theseus type type concept and it's all linked together by some fairly weak long-term memory and are we really the same person as we were a week ago a year ago we're far less philosophically concerned about that than we are by the micro level and the Meso Level but either way the solutions are the classic personal identity Solutions which have been gone over plenty of times if if you'll give us another five minutes before questions we'd love to share some more kind of speculative Thoughts with you um that's yeah getting a thumbs up wonderful in which case over to Andreas for some for some fun thought poking stuff well awesome awesome yeah I mean it's just kind of like this uh yeah deeper question and uh don't wait for me if my internet yeah runs out for a second but uh yeah I mean essentially how how does this influence for example yeah beliefs of personal identity uh next to the slide I think uh once you start seeing the universe as a field and with maybe like fleeting topological pockets on it that actually very significantly informs these classic

questions in philosophy like who are you and do you continue to exist from one moment to the other and India the ship of this ears like is there any kind of like metaphysical enduring ego so to speak and I actually think that each of these um views of personal identity could actually be reinterpreted in terms of claims about the field and field topology so something like closed individualism this is the belief that you start existing when you're born stop existing when you die or you know maybe you go to heaven but the point is that you're kind of a separate self that endures over time that is different than other selves here we would want to find some kind of a endearing sheath or enduring nested architecture that even though maybe each moment of awareness might be its own topological pocket maybe there's some kind of higher order topological Unity that does carry over a person's lifetime you know that would be a possible way of verifying in a actual you know significant physical way the sense of identity over time of course completely speculative would be a matter of actually finding it in the field but that's how you could I think still men this notion uh next slide then uh probably the most compatible and easy to justify view here would be empty individualism that every moment of experience is ontologically different and again here that would be possible because we would have these leading hard boundaries in the space-time Continuum and as a consequence you can yeah look at the Block universe and say okay there's a 4D pocket there there's a 4D pocket over there and wherever you see a 4D pocket that is going to be a subject of experience and you know any actual continuity between uh moments of experience are either just causal or kind of illusory constructed with the memory module as Chris was suggesting and then next slide and then finally open individualism there's this view that we are all one Consciousness we're all Christian at the bottom bottom layers so to speak well the way we could justify this within the topological account is that well actually and at the deepest anthropological level where all the field which has happened to be on a given point in time a segmentation of that field that prevents the information propagation with other regions of the field so in this case kind of the the topological segmentation would actually be more of an epistemological boundary then it would actually be an ontological boundary it's like the idea that like deep down actually we are the field it's just that how the field knows itself is constrained by its topology in in which case yeah it's another way of Steel meaning we're all one Consciousness in a very physically grounded way and yeah I think I think that's it unless I put anything else in the next slide oh yeah yeah just the last kind of like speculation here is uh well okay if you are this topological pocket in the field uh electromagnetic field um how can you in a sense take a look at that pocket and then reconstruct what the experience feels like and here perhaps it's like a fun

thought experiment that you could you could make is thinking about what it would be like to be an affordant trapped in a within a Hall of Mirrors uh there's this article uh I wrote yeah 2018 about how to replicate a hyperbolic space using just double mirrors and uh gradient index Optics uh and essentially within that structure that has those properties a photon would have no way of telling the fact that it is not in hyperbolic space in other words the index of refraction of a container would present itself as the actual geometry of the space from within so in our brain presumably like the actual the actual way in which the local field potentials are stitched together and the relative index of refraction could potentially give rise to really arbitrary possible internal geometrics where there's the illusion that you're in a space whether it's in hyperbolic space in spherical space on irregular geometry all of those could be emulated with these with these Paradigm and that may go a little bit uh a long way in actually explaining some of the most exotic experiences that people can have uh which usually would yeah challenge kind of theories of Consciousness here here's a way of us teaching them together to to explain them and I think yeah the last the slide I believe yeah so this is yeah just illustrating that point I think I think this is it yeah yeah and we can leave this up on the screen this is just where we would go next with it but this seems like a good point to stop and hear your thoughts awesome great presentations well Bernie would you like to give a first reflection and I'll write down some questions and look in the live chat yeah sure um I've been thinking about uh about similar stuff like this recently and um um when it's I the paper presents how the functional theories have the problem of arbitrariness uh so for example if marker blanket that can Marco blankets that could be the case when there's Marco blankets as ontological observers and I was thinking that maybe you need to find this with the market blanket Paradigm is that that on the physical level is implemented as as using this topological bucket is is the conscious marker blinking so it's how you were referring the marking black and solves the content problem while the the political boundary solves all the other problems like objects maybe at least yeah no that's something very quick to mention I mean obviously we're in the active inference Institute uh podcast I mean well absolutely one one thing to mention is that our perspective is not in itself necessary or is not necessarily sufficient actually to satisfy the you know that we have specified um however we actually think it's very very likely that topological pockets in the electromagnetic field do behave as marked with blankets and so it is in which you can actually interface these different paradigms where we would be saying that at the implementation level you know can then realize the immunization of free energy at the algorithmic level however we would be saying and maybe this would be a little

bit of a Twist or a non-trivial implication that if you were to simulate a you know hierarchically structured Markov blanket system in a computer it could very well be from a certain abstraction be minimizing free energy but it wouldn't have this it wouldn't be able to solve the boundary problem in other words it is sort of like there is a significant overlap but uh there are some slight places are very Divergence where you might say hey Markov blanket is actually not enough because it doesn't give you the implementation level explanation for how you get boundaries uh Chris if you want to add anything yeah yeah I mean just to just to agree with that I think this is the the if you go back to Markov three levels around kind of what substrate and we haven't discussed anything here about the algorithms that are actually taking place on or kind of within the brain that would ultimately get incorporated into the EM Fields clearly you need that stuff as well none of this goes makes any contribution to solving these more content problems of Consciousness or how we make inferences about the world how we respond to those what it does say is that for if there are algorithms that solve those problems and there are some good candidates out there including yours they need to be implemented or they need to reflect back to a field for it to become part of someone's Unified macro phenomenological experience if you did a some sort of feed forward in silico version of this or you did it you implemented the algorithm using rocks just lots and lots of rocks laid out forever like that wonderful XKCD example that that is not going to create a bound first-person perspective anywhere well a lot to say about that certainly Markov blankets and just to Define it on a Bayesian graph so the map not the territory the Markov blanket is the set of nodes upon which two other sets of nodes are conditionally independent and that's a purely topological statement it doesn't matter how far if there even is a spatial Association or temporal Association it doesn't matter the distance which is to say the geometry but rather the blanket is a purely topological statement and that's what among other things we've been exploring with Chris fields in the physics as information processing and this question about the topology of communication for bounded Observers and as we kind of alluded to with that content problem the topology of communication is to use a geometric metaphor it's orthogonal to the the content you could have two people on the phone sharing different things you could have a video file with different contents and so there there's really a lot of layers that come into play and so I think from an instrumentalist View I read your paper as a call for the introduction of topology into what is often only modeled as geometric a call for the integration of electromagnetism rather than a Reliance on one of the other of course as introduced from the outset of pluralism these are not exhaustive nor exclusive accounts and we don't know

what the scope of the possible intelligences and awarenesses are so from a research perspective if not Beyond why would we want to limit when we're not only interested in understanding the properties of any given system as we see it today any thoughts or I can read a question in the chat yeah that sounds great TurnKey tyranny writes how does topology show up phenomenologically yeah this is a excellent excellent question uh I'm happy to take it you want to go back to your notes you know you've got that page with notes in there very soon because this this is because I I've asked a very similar question of Andres and these sort of notes is what resulted in in his attempts to explain it to me so let me tell you yes very I mean this is very speculative and uh but but one one wonderful resource I think I think hopefully I think is wonderful very worth uh trying to get out at least we made a um series of guided meditations about uh phenomenology they're not it's not really kind of like particular spirituality it's more for the sake of exploring phenomenology uh with meditation the last one is called the phenomenology of ontology and that one helps you in a sense get a sense of I believe separating reality attacks your reality of course it's different to believe that the electromagnetic field or you're an electromagnetic but it feels like crystals or in a specific version of materialism or something like that so very different ontologies can give rise to very different feelings um and so there's definitely the worry that like well if you think a lot about the electromagnetic field aren't you may be just interpreting your Sensations within that Paradigm but like it's kind of like you're overfeeding on top of it um but uh you know in a sense I would argue that actually it's the other way around with something as generative and structured as electromagnetic field you can think in principle explain the phenomenology of any ontology like why does it feel like for example that you uh you can kind of like merge with the cosmos or something like that why does it feel like you can become one with everything in meditation let's say in various meditation States well with something like the topological explanation for uh the boundary problem the way we might interpret that is as the vanishing of internal boundaries and so there is a significant difference between a global boundary which is what we focused on the overall kind of the macroscopic division that says whether you're inside or outside your experience but something that the electromagnetic field also affords and gives us as kind of part of its Rich structure is the possibility of internal boundaries as well which would be not boundaries that encomp um that um wrap around a 4D object but rather lower dimensional boundaries like meeting points or convergence points of the field lines within the pocket and so in this account you know whenever you have like the sensation of solidity the feeling of well I am here you're there or there's a boundary between us

that boundary would essentially be an internal boundary would be a region of the field well the field lines converge or they they align um but it's only along a certain cross-section meaning that you know actually your experience is still fully unified you can still go from one point to the other across a perceived barrier because they're still part of the the whole experience they're still connected in other words with this kind of data structure we can arrive at kind of like a lot of um phenomenological details and trying to explain them from from that perspective uh and I think it's going to be a enormous program in in the long term you know involving physicists and mathematicians and exotic States Of Consciousness to say like okay that particular feature of experience that happens on certain substance in such a such meditation state how does that uh map into the electromagnetic field and I think we will very likely find a lot of non-trivial connections with that that sort of paradigm press any thoughts on that uh just to make the link to this uh to some of the the diagrams on the page which is if you've got these non-4d closed boundaries but other types of boundaries the other types of shapes Andres was talking about the drawings on the page are just examples of some of those but and and this is as Andrew said it's not really part of the solution but it's part of Imagine to be exploring where else it might go so if you've got you've got feature binding which is you know I'm seeing a cup in the cup is both blue and cup shaped that's feature binding and then the phenomenological binding is my whole experience being being unified you can imagine feature binding putting the blueness and the Cutlass together that might be something that happens along one of these types of internal boundaries it might not it might be that that can be done purely through informational associations where we're kind of agnostic on that but what's interesting about 4D topology is within it you can have lots of other complicated topologies provided they are not closed along the fourth dimension because then you've got you sort of created another little mini business awareness oh Bernie go ahead yeah I'll jump back to the other page uh yeah I wanted to say something else uh not related to this exact question so if you have ended it maybe I'll yeah well in that case actually I'll mention one last thing about the how electromagnetic field may be related to phenomenology uh a very specific thing that happens on psychedelics and especially something like DMT is that you may end up having uh different regions of your experience have the exact same texture and have the feeling of synchronization between them and when that happens they sort of become the same object just like spatially distributed um the way I interpret this is whenever the exact same wave equation applies to two regions of your experience and waves can propagate from one region to the other that actually feels like they're the same object so one kind of like natural

way in which yeah regions of your phenomenal world might be segmented out into phenomenal objects is by confining how waves can move within the world simulation and whenever you have a what feels like a bound object in your world simulation that I think is highly highly related to waves being able to smoothly travel within that object but not outside of that object some kind of wave confinement is happening I'll add in one more of course fun and speculative thought which is the aboutness the content of what this computation or what this process is unfolding the aboutness itself can be structure learning and so even for a fixed electromagnetic topology you can have variable informational topology case in point the Von Neumann computer the CPU architecture electromagnetically is not changing topologically we don't need to reroute the telephone connections for there to be different messages that spread and so there's a lot of richness in in the space you've you've sketched or surveyed because even for a fixed topology we might be able to do compositional um binding like blue and cup and then now we add in a dynamic multi-scale topological unfolding and then that may connect to different frequencies of for example measurements in the EEG which are really only picking up the synchrony in the outer shell which may actually play a role it's not like the skull matters you could have a skull removed it's not the scalp it's not your hair it's not necessarily the meningi but what is it well maybe it's something that's info topological that reflects kind of different regimes of attention that that actually does happen to correspond but it's also localized and that's that interplay with individuality and and Oneness that works like Wilbur's no boundary Point towards but then you got to come back from no boundary because that sentence has to end Bernie yeah I wanted to mention that in relation to the ongoing course on the physics and information processing that Chris Fields is hosting is one one construct that Chris Phillips is using in in his papers with Michael is the is the notion of topological Quantum neural networks and topological quantum computers so it might be that those are the the the information processing in in this kind of topological Maps as well that you've mentioned Daniel the uh grounding for that or how Michael Levin is talking about how build electricity is a glue for the various subsystems in the brain or uh based on I've seen some recent paper that was pretty late that that use there's this whole I I know I I fumbled recently which which I put the uh surprised up just recently that there is a whole whole field of people uh that are really trying uh to apply Quantum field theory on the brain and there was one paper on memory formation so how you Chris talked about uh memory in terms of the electromagnetic field that might be also great like pointer so I'm just putting out uh different people and papers that I think might be interconnectable with this word yeah one one

interesting thing you reflected it in your literature review is when people think about Quantum and cognitive systems the or or and other Frameworks often are considering Quantum as being electronic scale and therefore a quantum theory of cognition or Consciousness would entail some kind of relationship between the way that electrons and photons behave and then some difference that makes a difference at the biological level and in a way what we've been exploring with Chris fields and others is like a bypass where it's like no the quantum formalism Quantum topology Quantum information simply and plainly describes cognitive processes where like order matters now there may or may not be effects associated with differences that make a difference at the electronic scale but we don't need that in order to use like superposition and all of these different effects that empirically are the case in cognitive systems we're not waiting for some new mechanistic insight it'd be cool to find out but what you've presented is uh not waiting for a different kind of thing to exist so I'm curious what experiments or what are the unique experiments that you feel like would give a resounding feeling on what you're talking about yeah happy that they could uh take it on so uh uh one kind of like very awesome research lead and then I'll explain what a research Paradigm would look like but essentially a first of all like a very powerful lead is a meditative cessations so one of the most insane chapters in mastering the core teachings of the Buddha by Daniel Ingram is the one called the three doors where he describes the phenomenology of experiencing a peak meditation State that's called Association where you blink out of reality for a second or well for a sub second moment but you just blink out fully and then you get reconstituted and when you get reconcited it tends to be a very Blissful very rejuvenating type experience but the thing she highlights in that article is what it feels like to be really close to a cessation and it actually depends on what type of cessation it is there's roughly like three ways in which it can happen but the mind-blowing thing about that essay that chapter is he essentially describes different topologies of attention and awareness that may arise where in these moments close to Association he says well in one of those you become kind of a Taurus and the Taurus flips the inside and then outside and it twists around and then it cancels out and compresses and then you vanish and then there's like another variety where you become a cone and then you all of your attention gets concentrated in the point of the chlorine of the cone and then that twists around and then vanishes and so there's like this reliable you know very very uh exotic kind of like topological Transformations that can happen to your attention and awareness and our guess would be that well as a station you know an actual kind of full Vanishing might be related to the topological pocket that your brain is generating essentially opening up so that in

some maybe non-trivial way the actual boundary between inside and outside gets dissolved for a fraction of a second then it gets reconstituted but something really special happens in that that moment that allows for deep Insight or something along the lines or relieving a lot of internal stress maybe like a coronal mass ejection there might be something something going on going on at the topological level um now in terms of yeah kind of like further you know more sophisticated research paradigms the thing that we're looking into um there's a lot more to be unpacked and developed here but is what kind of electromagnetic resonant modes that can supervene on particular graphs you know whether we're talking about the connectum or whether we're talking about you know the the geometric eigen modes more more recent work where you model the entire cortex is that geometric uh geometric surface and then you apply neural field Theory to that well so like what kind like all of that works so far essentially what it focuses on is oscillations in the electric field without modeling the magnetic field so I'm actually you know I would suspect that many of the connective harmonics or eigen mode uh geometric eigen modes may be physically unrealizable simply because the magnetic field lines May collide with each other so like some of those may not be possible but one of the crazy fascinating unusual predictions that may come out of our whole Paradigm is that once we have a really good mapping of what resonant modes arise in the brain on different states of Consciousness we would predict that a when you're fully awake normally conscious there will be a very Central big topological Loop inside you that whenever you're let's say in deep Sleep um you might see a kind of mind dust you might see lots of tiny little little pockets and maybe when you're in an ecstatic you know experience like like um deep meditation it may do something else like the particular resonant modes there may create open field lines that don't form a loop so that is like the sort of crazy very novel type of prediction or at least research Paradigm that this enables that's that's really cool Chris anything out of that yeah I'm just uh draw the link between the fourth bullet here and what Andrews is setting out I don't think any of this is there are some toy bits of modeling you could do that could increase confidence one way or another but in terms of an actual experimental prediction you would need to combine this with some theory of where which bits of the brain are doing this so if Ward and Guevara are right that it's the thalamus that's doing this stuff we would expect to see around the thalamus a closed pocket and it should be possible in principle like this is probably billions of pounds worth of efforts but to model the thalmers in the surrounding bits enough and you can find out if there would actually be a closed pocket if there would be one in theory you can see roughly where it would be it moves around a bit but it's normally around here you then should

be able to I don't know if ethical consent is even possible for something like this but disrupt that with a very very targeted external em source think you know transcranial stimulation but just you know laser precise if we've now broken the bit around the thalamus but left everything else more or less untouched we predict if Warden Guevara plus us is correct we predict that your first person awareness would blink out like deep sleep like anesthesia like a meditative cessation if we are wrong or the combination of us is wrong then you'll still be conscious but you might just experience something really bloody weird but you'll still expect your contents have changed not the not the actual container of Consciousness hasn't blinked in and out and then you could apply that to other bits of the brain wherever you think there's I mean our other other prediction is that it must be enclosing the immediate memory model that we're unsure exactly where that is as well that that is also an outstanding research question so this is I mean it feels like doable in our lifetimes I think but none of this is you know give us 10 grand and we'll do it this summer none of it is that easy um yeah very interesting and neuroimaging already utilizes electromagnetic measurements and there are things that currently are described within a biomolecular paradigm like for example changes in effective connectivity between different regions of the brain well there's already a tract let's just say between certain brain regions and so maybe if there's um if the action potential is traveling down that 100 Lane Highway are lines then it's going to be like this Rippling em continuity whereas if they were totally disarrayed along that tract then you you wouldn't have those two regions contained but it'd be like kind of like two regions like blowing smoke rings at each other if they had enough synchrony to to push through the trouble and the friction such that there was a pocket that was able to you know you you squirt hot water into a cold pool there's ways to angle it and do it in all of the combinations of viscosity that it just diffuses instantly there's also ways where there would be a high temperature toroidal form that could pass through a colder media because it maintains the topological a lot of these things are again described anatomically or molecularly or electronically but the magnetic The Duality between electronics and magnetism is not contentious so there may be a lot of phenomena could be Revisited and I don't know whether that would provide indirect evidence with what we already know um or suggest new plausible experiments but it's an approach yeah welcome it and thanks Bernie as well for the pointers towards like Chris fields and Michael Levin I may come back to you for the uh the qft and on memory formation in case I've spotted some stuff on that but I'm not sure I've spotted the one it sounded like you're referring to so yeah it's an exciting time this sort of research yeah one last connection there with with active inference is you have

the electronic and the magnetic fields influencing each other that's the the basis of what philosophers might consider in terms of this Epi phenomenalism or not question and um if we had an active inference situation of two songbirds the active states of one become the sensory of the other so if they were just passing messages every single time step to each other then they would both be engaged in their own blanketed perception on the inbound and action selection on the outbounds and so they could basically be in this continual oscillatory relationship with causal efficacy both ways again or you could design a situation where the causal efficacy only flowed one way but the point is this is a topological approach that lets us explore that in a way that isn't only appealing to molecular or material phenomena and then leaving the whole question which is the whole question of the informational and the closure and The Binding leaving all those to be resolved by slicing and dicing with a scalpel which is not even a viable option as far as we already understand Bernie and then the authors what are your closing thoughts or how would you suggest that people get in the game or what should they do with their own topologies maybe I could note that Andres could expand on if he wants to know uh any link well yeah I mean I guess the at the PRI I mean I asked to be introduced by Mike in 2015 on principia qualia but the the correspondence respondents between uh yeah you know like the tri for many years now you know you can see a 2020 presentation about it there's many more content about it but the the main idea of the way it would actually you know work out here is the regions of the topologically closed uh field that behave like a smooth field May potentially correspond to a very Blissful experience so you know I'm very very optimistic of the potential of okay like something like a peak uh Blissful experience Big 5 mu DMT experience or something like that it would probably I suspect have like a very internally simplified topology where you know the geometry is as smooth as possible so that's kind of like an interesting yeah prediction or or angle and on all of these uh I think yeah maybe to to wrap up I would say that we you know would enormously welcome anybody who is uh technically Savvy for like uh doing interesting simulations of the electromagnetic field I mean we are we're doing ourselves some like really interesting experiments and in combination with uh classical neural networks but anybody's kind of like a you know the intersection between machine learning uh electromagnetic fields and uh simulations and um uh you know statistics machine learning essentially that that kind of intersection is something that could really potentially accelerate these research you know if you had the technical competency to yeah sort of like work on the analysis I was describing or finding the electromagnetic resonant modes of our graph and then analyzing its topology you know we

would love to collaborate that's like the sort of content we we dream of producing and yeah in the next year or two so thank you and thank you for having us it's been fantastic really really appreciate it I appreciate that Chris nothing substantial to that just just my thanks as well and you know I've I've scribbled down some thoughts about about the markup blankets and about some of those different research areas as well so you know we're getting stuff from this as well so yeah and hopefully this helps reach a few new people who might come up with either a brilliant idea why something we'd Miss and this this just can't work for somebody we want to hear that as well um but even more constructively ways that this could be tested or built out more gradually modeled yeah thank you for having us cool new ways of thinking within even one topological frame enables new action selection in the area and the era of video chatting we've made new topological relationships new connectivities that then disperse yet again and so it continues thank you all though thank you so much until next time thank you amazing stuff see ya bye